

Evaluating State Tracking in LLM-based Interactive Narrative Generation

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Abstract

Large Language Models (LLMs) often struggle to maintain narrative coherence and rule adherence over extended multi-turn interactions, primarily due to inherent limitations in state tracking. This paper investigates whether the integration of an external "Observer LLM" can enhance the consistency of story generation compared to a standalone LLM architecture. We evaluate four distinct story simulation frameworks: (1) a standalone LLM functioning as the sole Game Master with implicit state tracking (hiding numerical values from output), (2) a standalone LLM with explicit state tracking (revealing numerical changes and consumed rules after each turn), (3) a deterministic programmatic state tracker, and (4) a collaborative framework utilizing an auxiliary Observer LLM to monitor and rectify state deviations. Experimental results demonstrate that relying solely on a single LLM for hidden state maintenance is unreliable, yielding average state tracking accuracy scores of only 0.10 (strict rule enforcement) and 0.11 (flexible rule adaptation). In contrast, the integration of an Observer LLM significantly improves performance, achieving average accuracy scores of 0.58 and 0.61, respectively. These findings suggest that an external observational mechanism is essential for mitigating state degradation in long-horizon narrative reasoning tasks.

1 Introduction

Large language models (LLMs) have demonstrated strong capabilities in generating fluent and coherent narratives and supporting interactive dialogue. However, maintaining consistent representations of entities and world states across multiple interactions remains challenging (Papalampidi et al., 2022).

As a narrative progresses, previously established information and world states should remain consistent with subsequent descriptions and events. Failures to maintain such information can result in

contradictions, incorrect consequences, and inconsistencies in the evolving narrative (Ghosh et al., 2023).

This issue is particularly important in interactive narrative systems, where the generated response should remain consistent with the world state established by previous interactions. Unlike conventional text generation, an interactive Game Master must not only produce plausible narrative responses but also account for changes caused by player actions and previously established rules. For example, when an action consumes a limited resource, subsequent responses should reflect the updated resource state. Failure to track such changes can lead to contradictory outcomes and undermine the consistency of the interaction.

External memory, retrieval mechanisms, and structured representations such as knowledge graphs have been explored to provide LLMs with information beyond their immediate conversational context (Lewis et al., 2021; Gao et al., 2024; Gutiérrez et al., 2025). Such mechanisms can help preserve and retrieve relevant information that may otherwise be difficult for the model to maintain during generation.

However, in interactive narrative, the relevant information is not static: player actions continuously modify the underlying world state. Previous work on textual world models and generative agents has explored mechanisms for representing, retrieving, and maintaining information about evolving environments (Ammanabrolu and Riedl, 2019; Park et al., 2023). An LLM-based Game Master must therefore not only retrieve or remember previously established information, but also update the world state according to player actions and apply these changes consistently in subsequent interactions. The reliability of this dynamic state tracking remains an open empirical question.

To investigate this issue, we developed a prototype LLM-based Game Master within a customized

interactive narrative environment. The simulation scenario involves a spaceship crash, where players must survive in an abandoned space mining station and successfully send out a distress signal before resources run out.

The environment maintains explicit world-state variables, including resource levels and event conditions, while the Game Master is responsible for interpreting player actions and updating the narrative accordingly. Rather than proposing a complete solution to state-consistency problems, this study uses the environment as a controlled setting to examine how reliably an LLM can track and maintain dynamic world states across multiple interactions.

Research Question: *To what extent can an LLM-based Game Master maintain consistent world states and follow dynamically changing game rules across multiple interactions?*

2 Related Work

2.1 Interactive Narratives and Game Master Challenges

In domains characterized by complete information and explicit rules, such as chess and Go, artificial intelligence has already matched or surpassed human proficiency. However, open-ended environments driven by natural language, epitomized by tabletop role-playing games like Dungeons & Dragons (D&D), present the next frontier for AI (Louis and Sutton, 2018).

Evaluating AI systems within such highly open-ended settings—where dialogue must simultaneously reference background knowledge and adhere to complex rules—serves as a robust measure of linguistic and reasoning capabilities. Previous empirical analyses by Callison-Burch et al. (2022) revealed that when a Game Master LLM is forced to simultaneously manage narrative generation and rigorous state tracking, performance often degrades. This underscores the necessity of externalized support mechanisms to assist narrative models in preserving logical coherence and engaging plots.

2.2 Narrative Consistency and State Tracking

Recent studies have also investigated consistency in interactive narratives, including whether agents preserve previously established narrative commitments during interaction. (Ma et al., 2026) introduced the concept of narrative commitment preservation to evaluate whether generated events remain

consistent with previously established constraints. These studies highlight the importance of evaluating consistency not only at the level of individual responses, but also across the evolving state of an interaction.

Text-adventure games provide a particularly useful setting for this type of evaluation because they combine structured states, explicit rules, and sequential interactions. Such environments make it possible to evaluate whether an agent’s actions and responses remain consistent with the evolving game state. Recent work such as (Hu et al., 2026) further demonstrates the use of game environments for systematically evaluating LLM behavior in interactive settings. These studies motivate the use of controlled game environments for examining state tracking and rule adherence in interactive narrative systems.

2.3 External Memory and Knowledge Graph Integration

To mitigate these limitations, recent paradigms have increasingly incorporated external architectures to augment LLM decision-making. Park et al. (2023) introduced "Generative Agents" (the "Sandbox Village" framework), which simulate a memory stream consisting of observation, retrieval, and reflection mechanisms to provide external references for agent behavior.

Similarly, Ammanabrolu and Riedl (2019) proposed integrating dynamic knowledge graphs into text-adventure games, enabling narrative LLMs to more effectively select rational actions, continuously update plot contexts, and enhance the overall interpretability of the generated outputs. Building upon these insights, our work focuses on evaluating state tracking under controlled resource dynamics and dynamically introduced rules.

Existing studies have demonstrated the potential of external memory and structured knowledge for interactive narrative generation. However, our study focuses specifically on isolating and evaluating the state-tracking component under controlled resource dynamics and dynamically introduced rules.

3 Methodology

3.1 Simulation Environment and Narrative Setting

To evaluate narrative generation and state tracking, we implement a custom narrative simulation

platform powered by gpt-4o-mini (OpenAI et al., 2024) acting as the central Game Master LLM. The simulation scenario places the player in a survival crisis: following a spacecraft accident, the player must evacuate to an abandoned space mining station and strategically manage their actions under severe resource scarcity.

The simulation environment is governed by three core state attributes: power, oxygen, and stress. The game terminates under two failure conditions: resource depletion (i.e., when either power or oxygen drops to zero, indicating a total failure of life support systems) or psychological collapse (i.e., when stress exceeds a threshold of 10 points). To survive, the player must issue conversational actions to explore the environment, perform repairs, and ultimately transmit a distress signal.

3.2 Initial States and Predefined Consumption Rules

Inspired by resource-constrained evaluation frameworks in strategic environments (Yuan et al., 2025), we establish a controlled environment governed by three core states and a foundational set of consumption rules. Furthermore, we anticipate incorporating a Player LLM in future work to establish a fully automated game evaluation mechanism.

At the beginning of the simulation, the world state is initialized with 40 power, 20 oxygen, and 0 stress. Each player dialogue input represents one discrete simulation turn. State transitions are governed by a set of predefined deterministic consumption rules as follows:

1. **Power:** Each turn incurs a baseline cost of (-2); specific actions induce additional costs, such as area illumination (-1), operating a plasma welder (-10), and transmitting a distress signal (-5).
2. **Oxygen:** Each turn incurs a baseline cost of (-1); exploratory actions induce an additional cost of (-2).
3. **Stress:** Each turn incurs a baseline accumulation of (+1); actions performed in dark environments incur an additional (+2).

4 Experiments and Evaluation

4.1 Experimental Settings

To comprehensively evaluate state-tracking efficacy, we conduct two distinct experimental settings,

with each configuration executed across 5 independent trials:

1. **Constrained Setting:** The Game Master LLM is strictly prohibited from introducing consumption rules outside the predefined rulebook. The simulation is driven by a fixed sequence of 7 scripted player action turns.
2. **Open-Ended Setting:** The Game Master LLM is permitted to dynamically introduce new consumption rules not present in the predefined rulebook, provided that the impact of these rules is explicitly defined. These newly introduced rules are manually verified to determine whether they are properly factored into state tracking. This setting is driven by a fixed sequence of 8 scripted player action turns. According to manual trace calculations under static rules, the simulation should terminate at the 8th turn due to stress exceeding the maximum threshold. However, due to logging discrepancies, we unfortunately observed only a single instance of game termination independent of state values. This discrepancy prevented reliable evaluation of termination correctness and motivated us to focus the current analysis on per-turn state tracking rather than terminal-state prediction.

4.2 Evaluated Frameworks

We evaluate four framework configurations that differ in how world-state updates are maintained and represented. Table 1 summarizes the key architectural differences among these configurations. Figure 1 illustrates the overall information flow between the Player, Game Master LLM, state-tracking components, and persistent world state.

The first two configurations delegate state maintenance to the Game Master LLM, while the third and fourth configurations externalize state updates to dedicated components.

1. **Standalone LLM (Implicit):** The Game Master LLM is tasked with both narrative generation and implicit state tracking. Numerical changes are withheld until the game terminates. The required output format is structured as: [Environment][New Rules].
2. **Standalone LLM (Explicit):** Similar to the first framework, the Game Master LLM handles narrative generation, but performs explicit state tracking. The current state and

Table 1: Framework Comparison

Framework	State Update	State Representation	External Component
Implicit LLM	Game Master LLM	Hidden	None
Explicit LLM	Game Master LLM	Explicit	None
Programmatic	Python	External	Python Tracker
Observer LLM	Observer LLM	External	Observer LLM

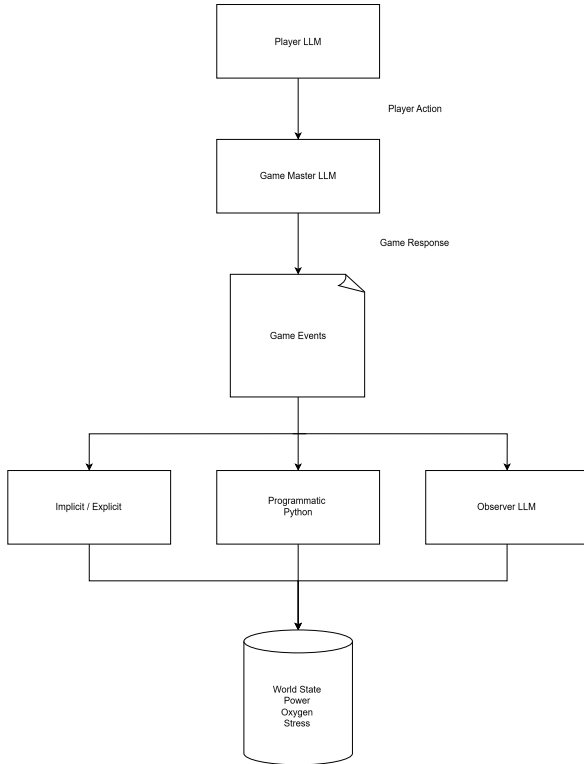


Figure 1: Information flow between the Player, Game Master LLM, state-tracking components, and persistent world state across the evaluated frameworks.

the specific rules applied are displayed at the end of each turn. The required output format is: [Environment][Current State][Applied Rules][New Rules].

3. Deterministic Programmatic State Tracker:

The Game Master LLM focuses solely on narrative generation, while state calculations are handled deterministically via Python code based on per-turn actions and fed back to the LLM to assist in subsequent story generation. The input/output formats are structured as follows: Player input comprises [Observer’s LastState Resolution][Player Action], while Game Master output consists of [Environment][New Rules].

4. Collaborative Framework with an Auxiliary Observer LLM:

Following the Game Master’s narrative generation, the dialogue history between the player and the Game Master is passed to an auxiliary Observer LLM. The Observer computes the state updates and feeds them back to the Game Master to guide narrative progression. The formatting is structured as follows: Player input comprises [Observer’s LastState Resolution][Player Action], Game Master output consists of [Environment][New Rules], and Observer output is formatted as [State][Applied Rules][New Rules].

4.3 Results and Analysis

We evaluate State Tracking Accuracy across five independent trials for each framework. State Tracking Accuracy measures the proportion of evaluated state updates that exactly match the ground-truth state after each turn. For the open-ended setting, we additionally evaluate Dynamic Rule Integration, which measures the proportion of newly introduced state-affecting rules that are correctly incorporated into subsequent state updates. Figure 2 summarizes state-tracking accuracy across the two experimental settings, while Table 2 reports dynamic rule

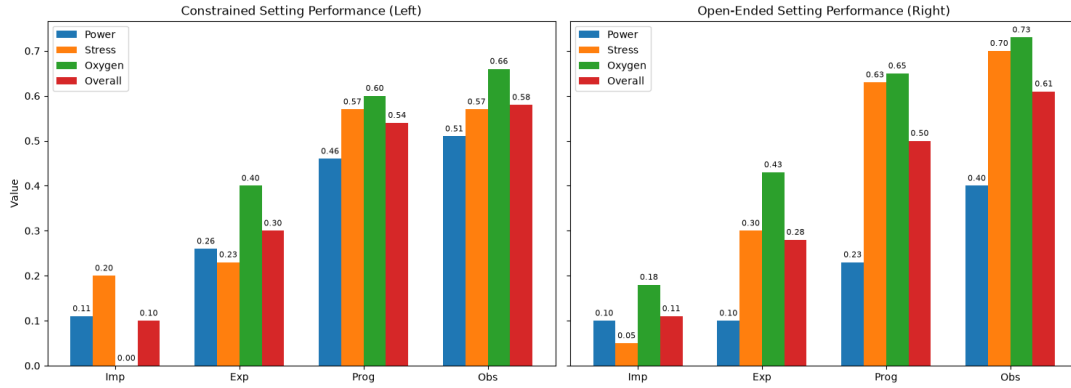


Figure 2: State-tracking accuracy across the four frameworks. (left) Constrained setting without dynamic rule creation. (right) Open-ended setting with dynamically introduced rules.

integration results.

4.3.1 Constrained Setting Performance

In the constrained setting, the Observer LLM achieves the highest overall accuracy (0.58), followed by the programmatic tracker (0.54), while the implicit and explicit LLM configurations achieve lower overall accuracies of 0.10 and 0.30, respectively.

Analysis: These results demonstrate that within this simulation platform, burdening a single LLM with both narrative generation and state maintenance fails to compute states accurately. Specifically, implicit state tracking yields near-zero reliability, particularly in maintaining oxygen metrics (0.00). While Framework (3) manages per-turn baseline consumption through deterministic code, it remains limited. In contrast, the auxiliary Observer LLM (Framework 4) successfully tracks state progression by reasoning over the dialogue history, achieving the highest baseline accuracy (0.58).

4.3.2 Open-Ended Setting Performance

For the open-ended setting, we additionally evaluate Dynamic Rule Integration, which measures the proportion of newly introduced state-affecting rules that are correctly incorporated into subsequent state updates.

Table 2 further summarizes the number of dynamically generated rules and the number of state-affecting rules that were correctly incorporated into subsequent state updates.

The Observer LLM correctly integrated all six state-affecting rules, whereas the implicit, explicit, and programmatic frameworks integrated six of eight, one of eight, and four of fifteen state-

affecting rules, respectively.

Analysis: Once the Game Master LLM is granted the freedom to dynamically introduce new rules, the Deterministic Programmatic State Tracker (Framework 3) struggles to incorporate these unprogrammed constraints, causing a sharp drop in Power tracking accuracy (plummeting from 0.46 down to 0.23). Interestingly, although Standalone LLM Explicit (Framework 2) outperforms its implicit counterpart in general state logging, it similarly fails to apply newly created rules to active state updates. In contrast, the Auxiliary Observer LLM (Framework 4) demonstrates superior adaptability and robustness; despite the introduction of dynamic rules, it maintains the highest overall accuracy (0.61) and successfully integrates 100% of the effective new rules (6 out of 6) into state updates. In our experimental setting, the Observer LLM achieved the highest state-tracking accuracy, particularly when dynamically introduced rules were present.

5 Conclusion & Future Work

5.1 Conclusion

In this paper, we presented a dialogue-based simulation testbed designed to evaluate the state-tracking performance of four distinct frameworks across two experimental settings. Although each experiment was conducted over 5 independent trials, these preliminary results indicate that the simulation platform can reveal substantial differences in state-tracking behavior across architectural configurations. Specifically, our findings confirm that burdening a standalone LLM with simultaneous narrative generation and state maintenance is unreliable. Furthermore, the Observer LLM demon-

Table 2: Dynamic Rule Integration

Framework	Correctly Integrated	State-affecting	Total New Rules
Implicit LLM	6	8	11
Explicit LLM	1	8	8
Programmatic	4	15	15
Observer LLM	6	6	6

strated greater adaptability to dynamically introduced rules than the deterministic tracker in our current setting.

Beyond quantitative metrics, our qualitative observations revealed intriguing failure modes in LLM narrative generation. For instance, in one trial set within the vacuum of an abandoned space mining station, the Game Master LLM generated a severe physical and logical contradiction by describing the player "opening a window, allowing outside oxygen to flood the station and secure a successful escape." This failure illustrates that state inconsistency is not limited to numerical bookkeeping; it can also manifest as violations of physical and causal constraints.

5.2 Future Work

Building upon these insights, we intend to pursue several directions for future research:

1. **Diverse Narrative Backgrounds:** Expanding the simulation testbed to different thematic scenarios to test framework generalizability.
2. **Dynamic Knowledge Graphs:** Integrating structured knowledge graphs to enhance the system’s external memory and long-term context retention.
3. **Automated Evaluation via Player LLMs:** Replacing scripted inputs with a Player LLM to establish a fully automated, end-to-end evaluation loop.
4. **Long-Horizon Dialogue Scaling:** Scaling the collaborative observer framework to handle extended multi-turn narrative sessions, thereby mitigating cumulative drift over long horizons.

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